

20 Poisson Approximation

- Poisson models are often used to model the distribution of random variables that count the number of “relatively rare” events that occur over a certain interval of time/in a certain location

Example 20.1. Let X be the number of home runs hit (in total by both teams) in a randomly selected Major League Baseball game. Assume that X has a Poisson(2.4) distribution.

1. In what ways is this like the Binomial situation?
2. In what ways is this NOT like the Binomial situation?
3. In what sense are home runs “relatively rare”?

- Binomial and Negative Binomial models have several restrictive assumptions that might not be satisfied in practice
- Poisson models are more flexible and are often used to model the distribution of random variables that count the number of “relatively rare” events that occur over a certain interval of time/in a certain location

Example 20.2. Recall the birthday problem: in a group of n people what is the probability that at least two have the same birthday? (Ignore multiple births and February 29 and assume that the other 365 days are all equally likely.) We will investigate this problem using Poisson

approximation. Imagine that we have a trial for each possible *pair* of people in the group, and let “success” indicate that the pair shares a birthday. Consider both a general n and $n = 30$.

1. How many trials are there?
2. Do the trials have the same probability of success? If so, what is it?
3. Are any *two* trials independent? To answer this questions, suppose that three people in the group are Abe, Bill, and Charles and consider any *two* of the trials that involve these three people.
4. Are any *three* trials independent? Consider the three trials that involve Abe, Bill, and Charles.
5. Let X be the number of *pairs* that share a birthday. Does X have a Binomial distribution?
6. If X has an approximate Poisson distribution, what would the parameter have to be? Compare this Poisson distribution with a simulation of X ; does it seem like a reasonable approximation?

7. Approximate the probability that at least two people share the same birthday. Compare to the theoretical value.

Poisson paradigm. Let $A_1, A_2, \dots, A_n$ be a collection of n events. Suppose event i occurs with marginal probability $p_i = P(A_i)$. Let $N = I_{A_1} + I_{A_2} + \dots + I_{A_n}$ be the random variable which counts the number of the events in the collection which occur. Suppose

- n is “large”,
- $p_1, \dots, p_n$ are “comparably small”, and
- the events $A_1, \dots, A_n$ are “not too dependent”,

Then N has an approximate Poisson distribution with parameter $E(N) = \sum_{i=1}^n p_i$.

Example 20.3. Use Poisson approximation to approximate the probability that at least *three* people in a group of n people share a birthday. How large does n need to be for the probability to be greater than 0.5?